GROWW AI DISCOVERY ENGINE: END-TO-END QUALITATIVE WORKFLOW

Target Strategic Metric: 12-Month First-Time Mutual Fund SIP Continuation Rate | Zero-Hallucination Evidence Pipeline

RAW INGESTED	QUALIFIED CORPUS	QUOTE EXACTNESS	EMERGENT THEMES	COMPETING HYPOTHESES	DOWNSTREAM TARGET
32547 Posts	3052 Records	100.0%	7 Clusters	14 Hypotheses	5-6 Interviews

STAGE 1 — Multi-Source Ingestion

- Play Store Reviews
- App Store Reviews
- r/IndiaInvestments
- YouTube Comments
- CSV Batch Imports

32547 Total Ingested

STAGE 2 — Cleaning & Qualification

- Regex PII Masking
- SHA-256 Deduplication
- Domain Relevance Filter
- Irrelevant Exclusion Log

3052 Qualified (100% PII Clean)

STAGE 3 — Structured LLM Extraction

- 21-Field Pydantic Schema
- 7-Stage Journey Enum
- Verbatim Quote Guardrail
- Unknown Demographics

100% Substring Match

STAGE 4 — TDD Quality & Golden Benchmark

- 20-Item Curated Benchmark
- 7 Agreement Dimensions
- 33 Logged AI Defects
- Prompt Refinements

Zero-Hallucination Gate

STAGE 5 — Pattern Discovery & Clustering

- Dense Text Embeddings
- UMAP 2D Projection
- HDBSCAN Density Cluster
- Counter-Evidence Mining

7 Emergent Themes

STAGE 6 — Hypotheses & 5-Pillar Ranking

- Competing Perspectives (A/B)
- Qualitative 5 Pillars
- Prevalence vs Impact
- 12M SIP Relevance

14 Falsifiable Hyps

STAGE 7 — Strategic Handoff & Interviews

- Streamlit Research UI
- Traceable Verbatim Quotes
- 1:1 Interview Guides
- Solution MVP Direction

5-6 Target User Sessions

TOP DISCOVERED OPPORTUNITY SPACES (PRIORITIZED FOR INTERVIEWS)

Ranked by 5-Pillar Qualitative Rubric (Prevalence, User Impact, 12M SIP Relevance, Quality, Product Addressability):

#1 Involuntary Churn: Bank Autopay Mandate & Deduct
Evidence: 548 quotes across 548 authors • Contradictory counter-evidence captured

#2 Over-Diversification & Portfolio Overlap Decisio
Evidence: 284 quotes across 284 authors • Contradictory counter-evidence captured

#3 Fear of Bank ECS Bounce Penalties During Tempora
Evidence: 200 quotes across 200 authors • Contradictory counter-evidence captured

CORE EPISTEMIC GUARDRAILS (§9.4)

- **Zero-Hallucination Quotes:** 100% exact substring match verified against raw text.
- **Epistemic Humility:** Demographics reported as 'unknown' unless stated explicitly.
- **Counter-Evidence Mining:** All themes retain resilient investors who continued.
- **Non-Conflation Rule:** Market drop distress kept distinct from actual SIP termination.
- **Target Handoff:** Part 1 supplies testable hypotheses for Part 2 user interviews.